Xianyu Chen

Research Scientist · Multimodal AI & Computer Vision · Gaze & Attention Modeling · Neural Signals
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SUMMARY

Research scientist in multimodal AI and human visual attention (ECCV 2024 Oral, top 2.3%; CVPR 2021, 2022, 2024; IEEE TPAMI). Ph.D. dissertation won the 2026 UMN Distinguished Dissertation Award. Now applies deep learning to peripheral neural signals at Fasikl.

EXPERIENCE

Fasikl Inc. (NeuroAI company) — Bloomington, MN
Research Scientist – AI — Oct 2025 – Present
Earlier: Research Engineer Intern (May – Aug 2023); Research Scientist Intern (May 2024 – Sep 2025; part-time from Aug 2024)

- Develop deep-learning methods for peripheral neural signals and closed-loop neuromodulation; co-inventor on two 2026 patent applications (stimulation control from neural and EMG signals; stimulation-parameter optimization).
- **SARM** (IEEE TBME 2026, co-author): a ConvLSTM that removes stimulation artifacts from nerve recordings, fine-tuned by reinforcement learning on motor-decoding rewards; it beat blanking and interpolation in benchtop tests.

University of Minnesota, Twin Cities — Minneapolis, MN
Ph.D. Researcher, Visual Information Processing Lab (advisor: Prof. Catherine Qi Zhao) — Sep 2019 – May 2024

- **GazeXplain** (ECCV 2024 Oral): predicts human scanpaths (where people look, in order) and describes in natural language what each fixation looks at; built a LLaVA-assisted, human-verified pipeline that labeled 86K+ scanpaths.
- **Gaze modeling** (CVPR 2021, 2024): built the first model of human scanpaths in visual question answering, trained with deep reinforcement learning; then personalized scanpath models to individual viewers with learned observer embeddings.
- **VisualHow** (CVPR 2022, co-first author; IEEE TPAMI 2024): co-led a dataset of 20K+ real-life problems with illustrated step-by-step solutions and human-attention labels; showed that supervising model attention with them improves solution prediction.

Tencent AI Lab · *Research Intern* — Summer 2019
Shenzhen Institutes of Advanced Technology, CAS · *Research Intern* (few-shot detection; IEEE TIP 2020) — Aug – Dec 2018

EDUCATION

University of Minnesota, Twin Cities — Minneapolis, MN
Ph.D. in Computer Science (GPA 4.0/4.0) — Dec 2025
Dissertation: *Building Human-Like Machine Intelligence: Advancing Attention by Modeling, Alignment, and Explainability*.

Sun Yat-sen University — Guangzhou, China
M.E. in Information and Communication Engineering (2018); B.E. in Communication Engineering (2015)

SELECTED PUBLICATIONS

First-author papers at ECCV, CVPR, IEEE TPAMI, IEEE TIP and IEEE TSP; 11 granted patents. Full list: chenxy99.github.io/publications.html

- GazeXplain: Learning to Predict Natural Language Explanations of Visual Scanpaths — ECCV 2024, Oral
- Beyond Average: Individualized Visual Scanpath Prediction — CVPR 2024
- Every Problem, Every Step, All in Focus: Learning to Solve Vision-Language Problems With Integrated Attention — IEEE TPAMI 2024
- VisualHow: Multimodal Problem Solving — CVPR 2022
- Predicting Human Scanpaths in Visual Question Answering — CVPR 2021

SKILLS

Machine learning: computer vision, vision-language models, explainable AI, eye-tracking (gaze) modeling, reinforcement-learning fine-tuning (policy gradient), transformers, LSTMs, graph neural networks, distillation, few-shot learning, biosignals
Tools: Python, PyTorch, TensorFlow, scikit-learn, NumPy, OpenCV, C/C++, MATLAB, Linux, Git

HONORS AND SERVICE

Honors: 2026 UMN Distinguished Dissertation Award (one of two university-wide); UMN CSE Graduate Fellowship (2019, 2020)
Service: Outstanding Reviewer, CVPR 2026 and WACV 2025; reviewer for CVPR, ICCV, ECCV, AAAI, IEEE TPAMI, and IEEE TIP